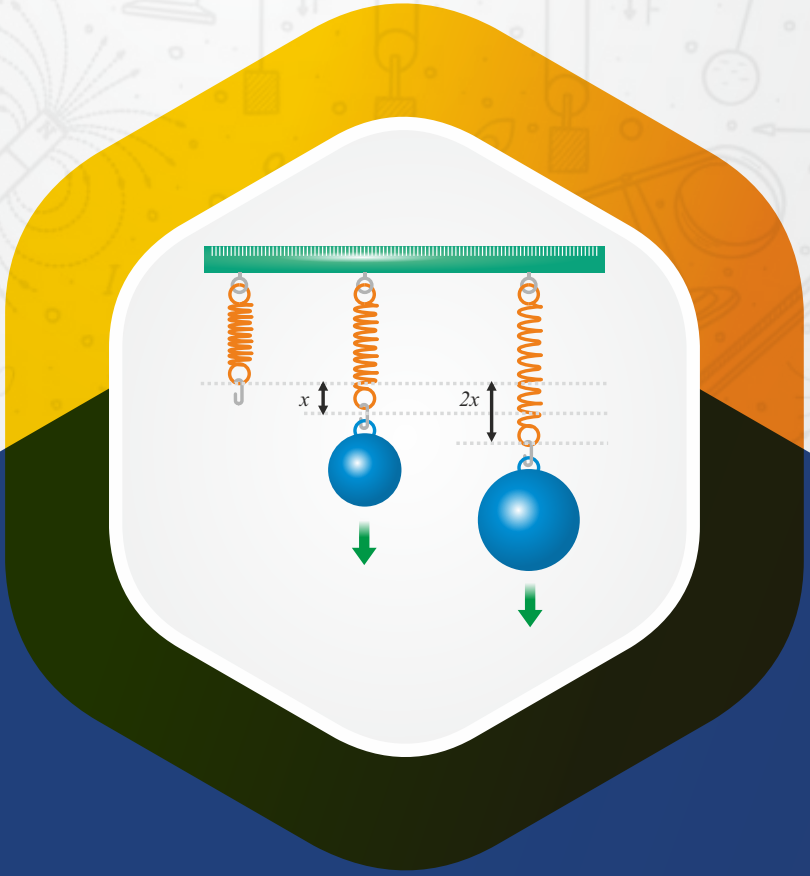
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Elasticity, Thermal Expansion, Calorimetry & Heat Transfer

(English Medium)

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ELASTICITY, THERMAL EXPANSION, CALORIMETRY & HEAT TRANSFER

KEY CONCEPT

ELASTICITY

STRESS

The internal restoring force acting per unit area of cross–section of the deformed body is called stress.

$$\text{Stress} = \frac{\text{Internal restoring force}}{\text{Area of cross section}} = \frac{F_{\text{internal}}}{A} = \frac{F_{\text{external}}}{A}$$

Stress depends on direction of force as well as direction of area of application so it is tensor.

SI UNIT : $N–m^{-2}$ **Dimensions :** $M^1 L^{-1} T^{-2}$

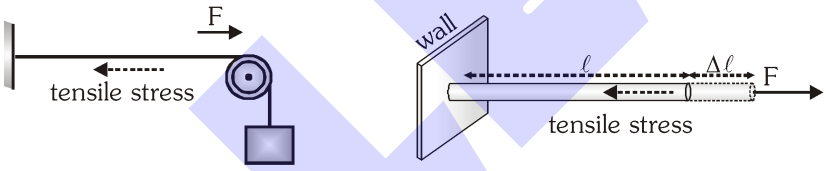
There are three types of stress :-

Longitudinal Stress :

When the stress is normal to the surface of body, then it is known as longitudinal stress. There are two types of longitudinal stress

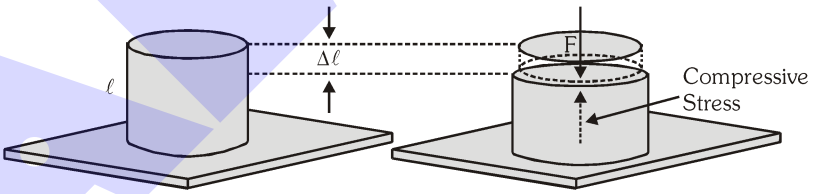
(a) Tensile Stress

The longitudinal stress, produced due to increase in length of a body, is defined as tensile stress.



(b) Compressive Stress

The longitudinal stress, produced due to decrease in length of a body, is defined as compressive stress.

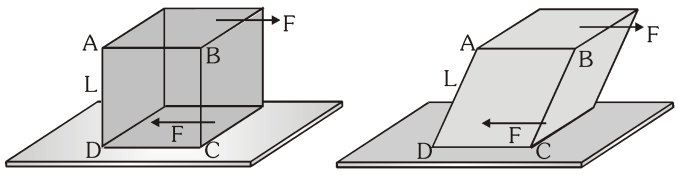


(c) Volume Stress

If equal normal forces are applied every one surface of a body, then it undergoes change in volume. The force opposing this change in volume per unit area is defined as volume stress.

(d) Tangential Stress or Shear Stress

When the stress is tangential or parallel to the surface of a body then it is known as shear stress. Due to this stress, the shape of the body changes or it gets twisted.



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STRAIN

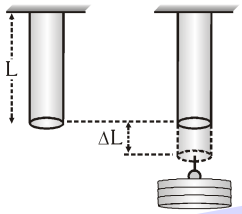
The ratio of change of any dimension to its original dimension is called strain.

$$\text{Strain} = \frac{\text{change in size of the body}}{\text{original size of the body}}$$

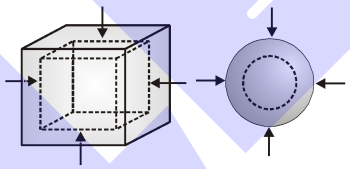
It is a unitless and dimensionless quantity.

There are three types of strain : Type of strain depends upon the directions of applied force.

Longitudinal strain = $\frac{\text{change in length of the body}}{\text{initial length of the body}} = \frac{\Delta L}{L}$

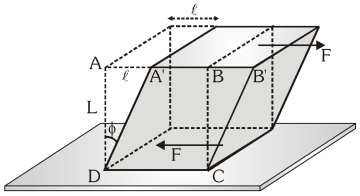


Volume strain= $\frac{\text{change in volume of the body}}{\text{original volume of the body}} = \frac{\Delta V}{V}$



Shear strain

When a deforming force is applied to a body parallel to its surface then its shape (not size) changes. The strain produced in this way is known as shear strain. The strain produced due to change of shape of the body is known as shear strain.



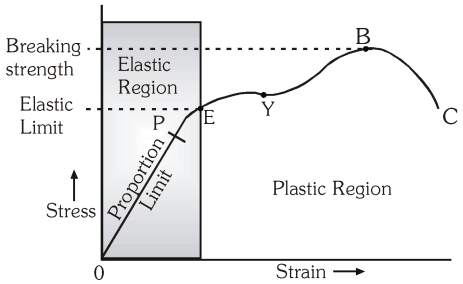
$$\tan\phi = \frac{\ell}{L} \text{ or } \phi = \frac{\ell}{L} = \frac{\text{displacement of upper face}}{\text{distance between two faces}}$$

When a material is under tensile stress restoring force are caused by intermolecular attraction while under compressive stress, the restoring force are due to intermolecular repulsion.

If the deforming force is inclined to the surface at an angle θ such that $\theta \neq 0$ and $\theta \neq 90^\circ$ then both tangential and normal stress are developed.

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STRESS – STRAIN GRAPH



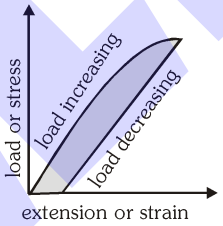
Proportion Limit : The limit in which Hooke's law is valid and stress is directly proportional to strain is called proportion limit.Stress $\propto$ Strain

Elastic limit : That maximum stress which on removing the deforming force makes the body to recover completely its original state.

Yield Point : The point beyond elastic limit, at which the length of wire starts increasing without increasing stress, is defined as the yield point.

Breaking Point : The position when the strain becomes so large that the wire breaks down at last, is called breaking point. At this position the stress acting in that wire is called breaking stress and strain is called breaking strain.

Elastic Hysteresis : The strain persists even when the stress is removed. This lagging behind of strain is called elastic hysteresis. This is the reason why the values of strain for same stress are different while increasing the load and while decreasing the load.



Breaking Stress : The stress required to cause actual facture of a material is called the breaking stress
Breaking stress = F/A

Breaking stress also measures the tensile strength.

Metals with small plastic deformation are called brittle.

Metals with large plastic deformation are called ductile.

Elasticity restoring forces are strictly conservative only when the elastic hysteresis is zero. i.e. the loading and unloading stress – strain curves are identical.

HOOKE'S LAW

If the deformation is small, the stress in a body is proportional to the corresponding strain, this fact is known as Hooke's Law. Within elastic limit : stress $\propto$ strain $\Rightarrow \frac{\text{stress}}{\text{strain}} = \text{constant}$

This constant is known as modulus of elasticity or coefficient of elasticity.

The modulus of elasticity depends only on the type of material used. It does not depend upon the value of stress and strain.

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YOUNG'S MODULUS OF ELASTICITY 'Y'

Within elastic limit the ratio of longitudinal stress and longitudinal strain is called Young's modulus of elasticity.

$$Y = \frac{\text{longitudinal stress}}{\text{longitudinal strain}} = \frac{F/A}{\ell/L} = \frac{FL}{\ell A}$$

BULK MODULUS OF ELASTICITY 'K' or 'B'

Within elastic limit the ratio of the volume stress and the volume strain is called bulk modulus of elasticity.

$$K \text{ or } B = \frac{\text{volume stress}}{\text{volume strain}} = \frac{F/A}{\frac{-\Delta V}{V}} = \frac{\Delta P}{\frac{-\Delta V}{V}}$$

The minus sign indicates a decrease in volume with an increase in stress.

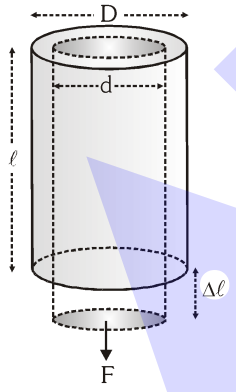
MODULUS OF RIGIDITY 'η'

Within elastic limit the ratio of shearing stress and shearing strain is called modulus of rigidity of a material.

$$\eta = \frac{\text{shearing stress}}{\text{Shearing strain}} = \left(\frac{\frac{F_{\text{tangential}}}{A}}{\phi}\right) = \frac{F_{\text{tangential}}}{A\phi}$$

Note : Angle of shear 'ϕ' always taking in radian

Poisson's Ratio (σ)



In elastic limit, the ratio of lateral strain and longitudinal strain is called Poisson's ratio.

$$\sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{\beta}{\alpha};\ \beta = \frac{-\Delta D}{D} = \frac{d-D}{D} \ \&\ \alpha = \frac{\Delta L}{L}$$

EFFECT OF TEMPERATURE ON ELASTICITY

When temperature is increased then due to weakness of inter molecular force the elastic properties in general decreases i.e. elastic constant decreases.

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THERMAL EXPANSION

Definition of Heat :

Heat is a form of energy which is transferred between a system and its surrounding as a result of temperature difference only.

Thermal Expansion : Expansion due to increase in temperature.

1. Type of thermal expansion

	Coefficient of expansion	**For temperature change Δt change in**	
(i) Linear	$\alpha = \underset{\Delta t\to 0}{\text{Lim}} \frac{1}{l_0}\frac{\Delta l}{\Delta t}$	length	$\Delta l = l_0 \alpha \Delta t$
(ii) Superficial	$\beta = \underset{\Delta t\to 0}{\text{Lim}} \frac{1}{A_0}\frac{\Delta A}{\Delta t}$	Area	$\Delta A = A_0\beta\Delta t$
(iii) Volume	$\gamma = \underset{\Delta t\to 0}{\text{Lim}} \frac{1}{V_0}\frac{\Delta V}{\Delta t}$	volume	$\Delta V = V_0\gamma\Delta t$

(a) For isotropic solids $\alpha_1 = \alpha_2 = \alpha_3 = \alpha$ (let)
so $\beta = 2\alpha$ and $\gamma = 3\alpha$

(b) For anisotropic solids $\beta = \alpha_1 + \alpha_2$ and $\gamma = \alpha_1 + \alpha_2 + \alpha_3$
Here α_1 , α_2 and α_3 are coefficient of linear expansion in X , Y and Z directions.

2. Variation in density : With increase of temperature volume increases so density decreases and vice-versa.

$$d = \frac{d_0}{(1+\gamma\Delta t)}$$

For solids values of γ are generally small so we can write $d = d_0(1-\gamma\Delta t)$ (using binomial expansion)

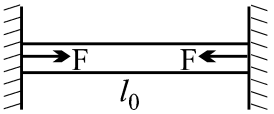
Note :

(i) γ for liquids are in order of 10^{-3}

(ii) For water density increases from 0 to 4°C so γ is –ve (0 to 4° C) and for 4° C to higher temperature γ is +ve. At 4° C density is maximum.

3. Thermal Stress : A rod of length l_0 is clamped between two fixed walls with distance l_0. If temperature is changed by amount Δt then

stress = $\frac{F}{A}$ (area assumed to be constant)



strain = $\frac{\Delta l}{l_0}$

so, $Y = \frac{F/A}{\Delta l/l_0} = \frac{Fl_0}{A\Delta l} = \frac{F}{A\alpha\Delta t}$

or $F = Y A \alpha \Delta t$

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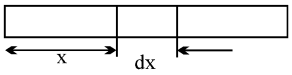
4. If α is not constant

(i) (α varies with distance) Let α = ax+b

$$\text{Total expansion} = \int expansion\ of\ length\ dx = \int_0^1 (ax+b)dx\,\Delta t$$

(ii) (α varies with tempearture)

Let α = f (T)



$$\Delta l = \int_{T_1}^{T_2} \alpha l_0\, dT$$

Caution : If α is in °C then put T_1 and T_2 in °C. Similarly if α is in K then put T_1 and T_2 in K.

CALORIMETRY

Quantity of heat transfered and specific heat

The amount of heat needed to incerase the temperature of 1 gm of water from 14.5° C to 15.5°C at STP is 1 calorie

dQ = mcdT

$$Q = m\int_{T_1}^{T_2} C\,dT$$ (be careful about unit of temperature, use units according to the given units of C)

Heat transfer in phase change

Q = mL L = latent heat of substance in cal/ gm/ °C or in Kcal/ kg/ °C

L_{ice} = 80 cal/ gm for ice

L_{steam} = 540 cal/ gm

HEAT-TRANSFER

(A) Conduction : Due to vibration and collision of medium particles.

(i) Steady State : In this state heat absorption stops and temperature gradient throughout the rod becomes constant i.e. $\frac{dT}{dx}$ = constant.

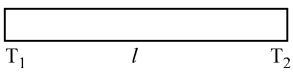
(ii) Before steady state : Temperature of rod at any point changes

Note : If specific heat of any substance is zero, it can be considered always in steady state.

1. Ohm's law for Thermal Conduction in Steady State :

Let the two ends of rod of length l is maintained at temp T_1 and T_2($T_1 > T_2$)

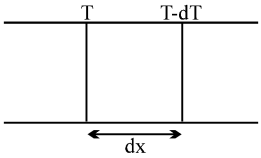
$$\text{Thermal current } \frac{dQ}{dT} = \frac{T_1 - T_2}{R_{Th}}$$



Where thermal resistance $R_{Th} = \frac{1}{K}\frac{1}{A}$

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2. Differential form of Ohm's Law



$\frac{dQ}{dT} = KA\frac{dT}{dx}$; $\frac{dT}{dx}$ = temperature gradient

(A) Convection : Heat transfer due to movement of medium particles.

(B) Radiation: Every body radiates electromagnetic radiation of all possible wavelength at all temp >0K.

(i) Stefan's Law : Rate of heat emitted by a body at temp T K from per unit area $E = \sigma T^4$ J/sec/m^2

Radiation power $\frac{dQ}{dT} = P = \sigma AT^4$ watt

If a body is placed in a surrounding of temperature T_S : $\frac{dQ}{dT} = \sigma A\,(T^4 - T_s^4)$ valid only for black body

Emissivity or emmisive power (e) = $\frac{\text{heat from general body}}{\text{heat from black body}}$

If temp of body falls by dT in time dt : $\frac{dT}{dt} = \frac{eA\sigma}{mS}(T^4 - T_s^4)$ (dT/dt = rate of cooling)

(ii) Newton's law of cooling

If temperature difference of body with surrounding is small i.e. $\Delta T << T_s$ then $\frac{dT}{dt} \propto (T - T_s)$

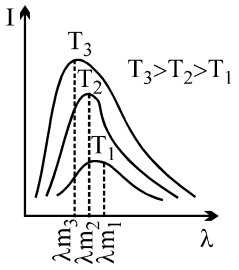
(iii) General form of Newtons law of cooling

If a body cools from T_1 to T_2 in time δt

$\frac{T_1 - T_2}{\delta t} = \frac{K}{mS}\left(\frac{T_1 + T_2}{2} - T_s\right)$ (used generally in objective questions)

$\frac{dT}{dt} = \frac{K}{mS}(T - T_s)$ (for better results use this generally in subjective)

(iv) Wein's black body radiation



At every temperature (> 0K) a body radiates energy radiations of all wavelengths.

According to Wein's displacement law if the wavelength corresponding to maximum energy is $\lambda_{m.}$ then $\lambda_m T = b$

where

b = is a constant (Wein's constant)

T = temperature of body

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